IMPERIAL COLLEGE LONDON

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING
EXAMINATIONS 2024

COMPUTATIONAL SENSING AND IMAGING

Monday, 20 May 10:00 am

Time allowed: 3:00 hours

There are ONE questions on this paper.

Answer ALL questions.

Any special instructions for invigilators and information for candidates are on page 1.

Examiners responsible First Marker(s) : Bhandari,A.

Second Marker(s) :

Other Dept

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Department of Electronic and Electrical Engineering

Examinations 2024

Computational Sensing and Imaging

MSc and EEE/EIE PART IV

May, 2024 • Exam Duration: Three Hours

All Questions are Compulsory

Where unavoidable, make reasonable assumptions for your solutions. Explain your assumptions in the answer script.

Markers: Ayush Bhandari and Tania Stathaki.

Questions [100 Marks]

1) What is Computational Sensing?
How is it different from signal or image processing? **[2 Marks]**

2) Provide 2 examples of computational sensing based systems. **[2 Marks]**

3) Consider a frequency estimation problem:

$$f(x) = c_1\alpha^x + c_2\beta^x, \qquad x = 0, 1, 2, 3.$$

Establish the following relation,

$$\begin{bmatrix} f(0) & f(1) \\ f(1) & f(2) \end{bmatrix} \begin{bmatrix} \alpha\beta \\ -\alpha-\beta \end{bmatrix} = -\begin{bmatrix} f(2) \\ f(3) \end{bmatrix}$$

Hint: You can see that this is related to Prony's method extensively covered in the class. 1) Use the annihilation polynomial in Prony's method: $h(z) = (z-\alpha)(z-\beta)$. Note that, $(f * h)[n] = 0$ where $\{h[n]\}_{n=0}^{2} = \{\lambda_0, \lambda_1, 1\}$, and 2) Utilize the fact that $h(\alpha) = h(\beta) = 0$. **[8 Marks]**

4) Consider the following constrained least squares problem of the form:

$$\min_{\mathbf{x}} \|\mathbf{y} - \mathbf{H}\mathbf{x}\|_2^2, \quad \text{such that} \quad \mathbf{C}\mathbf{x} = \mathbf{b}. \tag{1}$$

Show the optimal solution is:

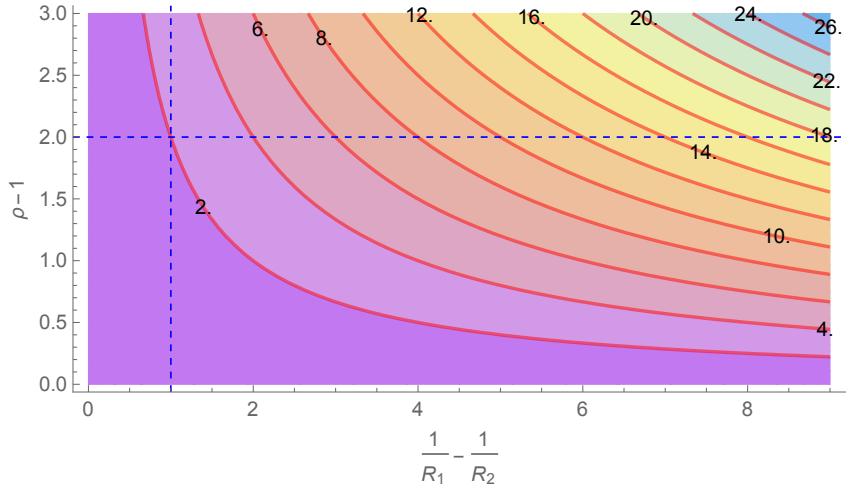
$$\mathbf{x}^* = (\mathbf{H}^\top\mathbf{H})^{-1}\left(\mathbf{H}^\top\mathbf{y} - \mathbf{C}^\top(\mathbf{C}(\mathbf{H}^\top\mathbf{H})^{-1}\mathbf{C}^\top)^{-1}(\mathbf{C}(\mathbf{H}^\top\mathbf{H})^{-1}\mathbf{H}^\top\mathbf{y} - \mathbf{b})\right).$$

[8 Marks]

5) Let $\rho = n_2/n_1$, the ratio of refractive indices.
The well known Lens Maker's equation is given by,

$$(\rho - 1)\left(\frac{1}{R_1} - \frac{1}{R_2}\right) = \frac{1}{f}$$

and for a certain imaging system, the operation range is described in the figure below.



(a) Explain the solid red lines in this figure. **[2 Marks]**

(b) Also, given materials such that $\rho \geqslant 3$, discuss the constraints on $\{R_1, R_2\}$ so that an effective focal length of half can be attained by the system. **[3 Marks]**

6) Sparse recovery beyond soft thresholding **[15 Marks]**

In the context of sparse recovery, the cost function

$$\mathcal{C}_\lambda(\mathbf{f}) = \|\mathbf{g} - \mathbf{f}\|_2^2 + \lambda \|\mathbf{f}\|_{\ell_1} = \sum_{n=1}^{N} (g_n - f_n)^2 + \lambda |f_n|$$

can be generalized in that the ℓ_1–norm can be replaced by a pointwise nonlinearity defined by function ξ, leading to

$$\mathcal{C}_\lambda(\mathbf{f}) = \|\mathbf{g} - \mathbf{f}\|_2^2 + \lambda \xi(\mathbf{f})$$

provided that ξ is differentiable. The core idea is to use a ξ that mimics the ℓ_1–norm.

(a) Explain why we need ξ to be differentiable.

(b) Suppose that we define

$$\xi(\mathbf{f}) = \frac{1}{\alpha} \log(1 + \alpha |\mathbf{f}|), \quad \alpha > 0.$$

For one-dimensional and two-dimensional $\mathbf{f}$ plot $\xi(\mathbf{f})$ and compare it with the ℓ_1–norm of $\mathbf{f}$.

In analogy to the minimizer given by, $\mathbf{f}^\star = \text{soft}_\lambda(\mathbf{g})$, what is the minimizer of the cost function with $\xi(\mathbf{f})$ defined as previously?

7) (a) What is the qualitative and mathematical definition of Plenoptic Function? Identify the different variables of the function. **[2 Marks]**

(b) Based on (a), explain (mathematically) how a smart-phone camera produces a 2D photograph from a plenoptic function. **[3 Marks]**

(c) Map each parameter in the plenoptic function (a) to an imaging modality. **[5 Marks]**

8) Noise Shaping Based Reconstruction.

Given a bandlimited function $f(t)$, in one-bit sampling, the samples $q[n] \in \{-1, 1\}$ are given by,

$$\begin{cases} q[n] &= \text{sgn}(u[n-1] + f[n]) \\ u[n] &= u[n-1] + f[n] - q[n] \end{cases}.$$

(a) Re-write the one-bit samples as a single equation in terms of f and u. **[3 Marks]**

(b) From above, derive an equivalent expression in the Fourier domain. **[2 Marks]**

(c) Show how an approach similar to Shannon's interpolation formula (low-pass filtering) can be used for reconstruction from one-bit samples. **[5 Marks]**

9) In a certain time-of-flight system, the emitted signal is give by $p(t) = e^{-\frac{t^2}{2\sigma^2}}$ (Gaussian Excitation). Suppose the scene response function consists of a single reflection with reflectivity Γ and the object distance $d = \frac{ct_0}{2}$ where c is the light speed.

Use $\int_{\mathbb{R}} e^{-\frac{t^2}{2\sigma^2}} dt = \sqrt{2\pi}\sigma$ to simplify your computations.

(a) Write down the expression of the reflected signal $r(t)$. **[1 Marks]**

(b) Write down the cross-correlation signal $f(\tau) = \int r(t)\, p(t+\tau) dt$.

[4 Marks]

(c) Explain why the (above) cross-correlation technique is sub-optimal for resolving close-by reflections.

[4 Marks]

(d) For a single reflection, explain how sparse deconvolution method can be used for depth recovery. **[3 Marks]**

10) Indicate which of the following statements that are true.

Real-world signals are

(a) analog in amplitude,

(b) quantized in amplitude,

(c) vary continuously in time,

(d) are discrete in time.

(Multiple answers are possible.) **[2 Marks]**

11) How is a discrete signal scaled so that the sampling frequency satisfies $f_s = 1$? If the original analog signal has bandwidth Ω rad/s the scaled samples represent samples from an analog signal of what bandwidth? **[2 Marks]**

12) In the context of signal/image denoising, people have developed filtering and thresholding techniques. Suppose a noisy signal $y[n]$ is given by $y[n] = x[n] + w[n], n = 0, \cdots, N-1$ where the original signal $x[n]$ is a finite sum of complex exponentials $x[n] = \sum_{k=0}^{K-1} c_k e^{\jmath \omega_k n}$ and $\max_k |\omega_k| \leqslant \Omega$ and where $w[n]$ is additive white Gaussian noise. Noting that $x[n]$ is sparse in Fourier domain, then the denoising problem can be formulated as

$$\min_{\widehat{\mathbf{x}}} \|\widehat{\mathbf{y}} - \widehat{\mathbf{x}}\|_2^2 + \lambda \|\widehat{\mathbf{x}}\|_{\ell_1}. \tag{2}$$

where $\widehat{\mathbf{x}}$ and $\widehat{\mathbf{x}}$ are the DFT of $\mathbf{x}$ and $\mathbf{y}$, respectively. Utilizing the soft-thresholding operator, write down the sparse recovery algorithm to solve (2) and explain the performance difference compared to the low-pass filtering methods.

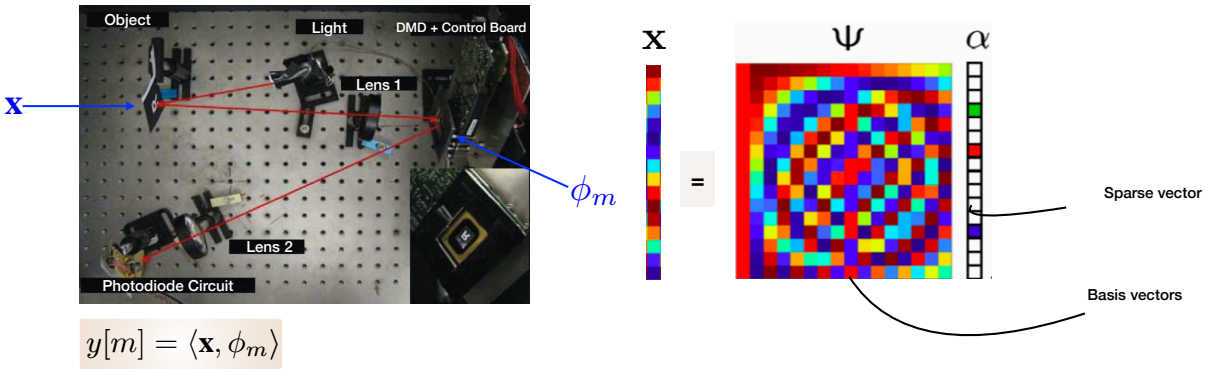
[5 Marks]

13) Based on the diagram shown in Fig. 1 ($m = 0, \cdots, M-1$), write down the forward model of the single-pixel imaging. Show that the single-pixel imaging problem can be solved using compressive sensing methods. **[5 Marks]**

14) What is the main difference between multi-spectral and hyper-spectral imaging?

[3 Marks]

15) Given two examples of imaging techniques with non-visible frequencies. **[3 Marks]**



$y[m] = \langle \mathbf{x}, \phi_m \rangle$

Fig. 1: Illustration of the single pixel imaging.

16) Compared to the conventional 2D photography, what is the significant feature of time-resolved imaging?

[2 Marks]

Furthermore, why it is difficult to achieve both temporal and spatial resolution in time-resolved imaging technologies?

[2 Marks]

17) Consider a linear regression problem $\min_c \|\mathbf{y} - c\mathbf{x}\|_2^2$ where $\mathbf{x}, \mathbf{y} \in \mathbb{Z}^N$.

Show that the minimum least-squares error is given by

$$\|\mathbf{y}\|_2^2 - \frac{|\mathbf{x}^{\mathsf{H}}\mathbf{y}|^2}{\|\mathbf{x}\|_2^2}.$$

[4 Marks]